Salbutamol

Mechanism of action

- Salbutamol is a bronchodilator. It is an agonist (stimulator) of beta-2 receptors.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Bronchospasm secondary to asthma or COPD.
- ✓ Prominent bronchospasm secondary to airway burns, smoke inhalation or chest infection.
- ✓ Release syndrome following crush injury.
- ✓ Known or suspected hyperkalaemia with ECG changes.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safe and may be administered if indicated.

Dosage

- MDI: One puff into the MDI and spacer, followed by six breaths. Repeat this six times.
- Nebulised for bronchospasm: 5 mg for adults and children, repeated as required. The initial dose is combined with 0.5 mg ipratropium, but subsequent doses are not.
- Nebulised for release syndrome following crush injury or known or suspected hyperkalaemia: 5 mg for adults and children, repeat as required.

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Administration

- MDI: Administer one puff of salbutamol using an MDI and spacer, followed by six breaths. Repeat this six times.
- Nebulised: Administer nebulised undiluted using:
  - Air as the driving gas for COPD.
  - Oxygen as the driving gas for all other indications.

Common adverse effects

- Tremor.
- Tachycardia.

Usual onset of effect

- 2-5 minutes.

Usual duration of effect

- 1-2 hours.

Usual preparation

- Metered dose inhaler delivering 100 mcg doses per puff.
- Ampoule containing 5 mg in 2.5 ml.
- Ampoule containing 2.5 mg in 2.5 ml.

Pharmacokinetics

- Only a small amount of the nebulised dose is absorbed, with most of the dose being nebulised to the atmosphere. The inhaled salbutamol is absorbed through the lungs and some is swallowed.
- Salbutamol is metabolised in the liver and excreted in urine.
- There are no significant effects from liver or kidney impairment on acute administration.

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Common interactions

- Salbutamol will be less effective in the presence of a beta-blocker, with the reduction in effect being most pronounced with a non-selective beta-blocker such as propranolol.

Additional information

- Salbutamol does not have a significant role in the treatment of bronchospasm as a result of smoke, toxic gas inhalation or chest infection. However, it may be administered if bronchospasm is prominent.

More Information

Version history

11 September 2023 1.0.4
Update: Indications, dosage and administration information updated.

22 November 2022 1.0.1
Update: Aligned with asthma and COPD CPGs.